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2026-03-21

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1: 3
2: 3
3: ENSO 5
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. 6

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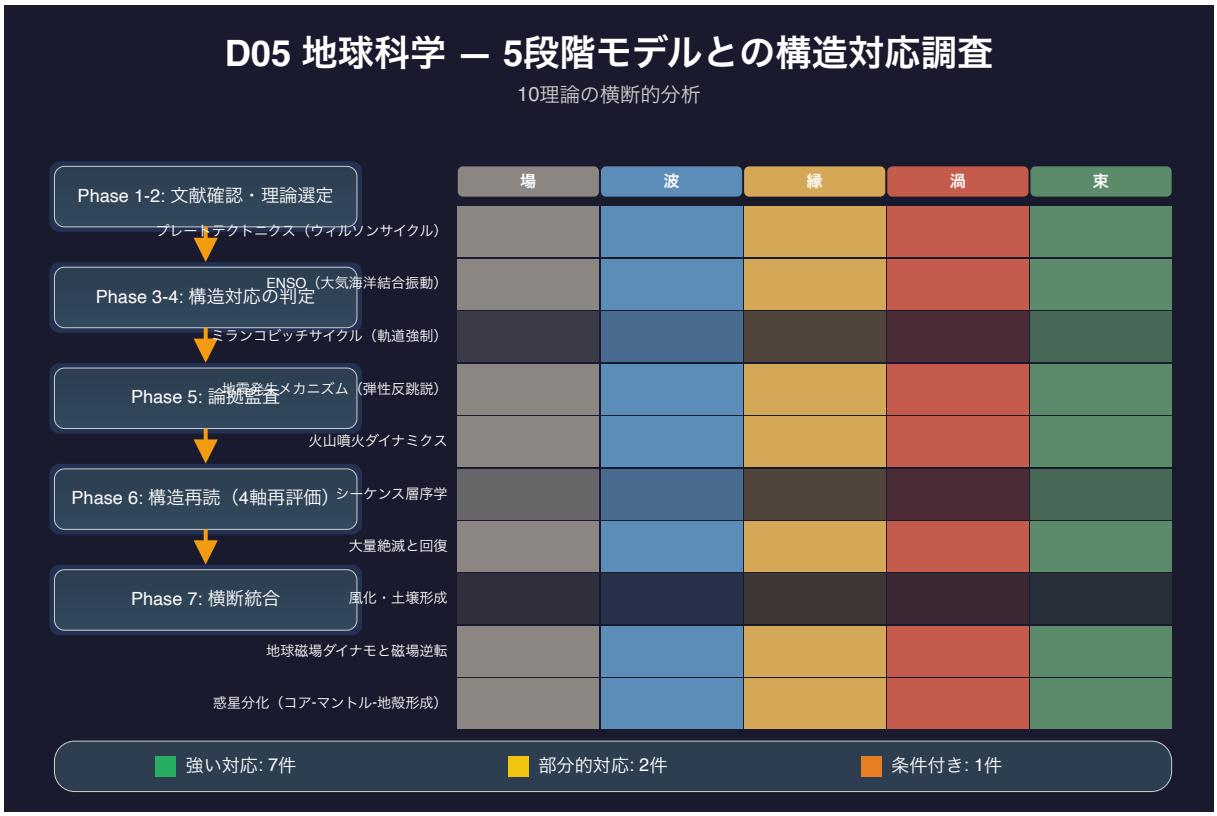
- : 14
- : 5
- : 7 3

5

#	/	5
1	Wilson, Morgan, Le Pichon, 1960	

#	/		5
2		Conrad and Lithgow-Bertelloni, 2000	
3	ENSO Bjerknes feedback	Bjerknes, Cane and Zebiak, 1969-1985	
4	Walker	Walker, 1920	5
5		Milankovitch, Hays, Imbrie, 20	
6	100kyr	Lisiecki and Raymo, 2005	
7		Reid, 1910	
8		Dieterich, Ruina, 1979-1983	
9		Cashman and Sparks, Gudmundsson, 1980-2010	
10		Vail, Posamentier, 1970-1980	
11		Raup, Sepkoski, Erwin, 1980-2000	
12		Goldich, Jenny, 1938-1941	
13		Elsasser, Glatzmaier and Roberts, 1946-1995	failed
14		Kleine, Solomatov, Elkins-Tanton, 2000	5



1: Domain —

1:

- Elkins-Tanton 2012 Hf-W Kleine et al. 2002 Solomatov 2000; Taylor and McLennan 1985
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2:

- Morgan 1968; Le Pichon 1968 → → → → → Wilson 1966; Dewey and Bird 1970 failed rift
- failed rift
- 5
- failed rift
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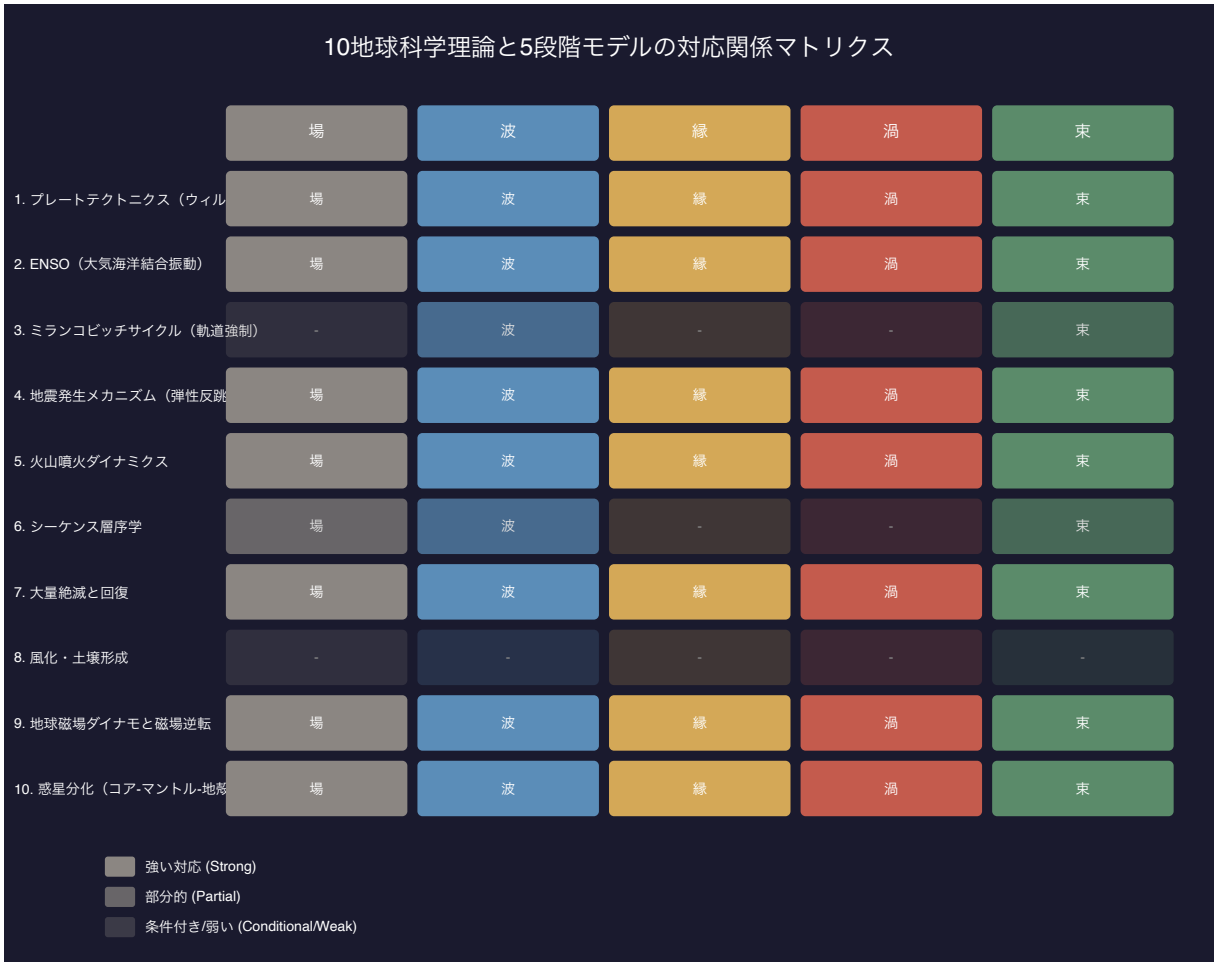
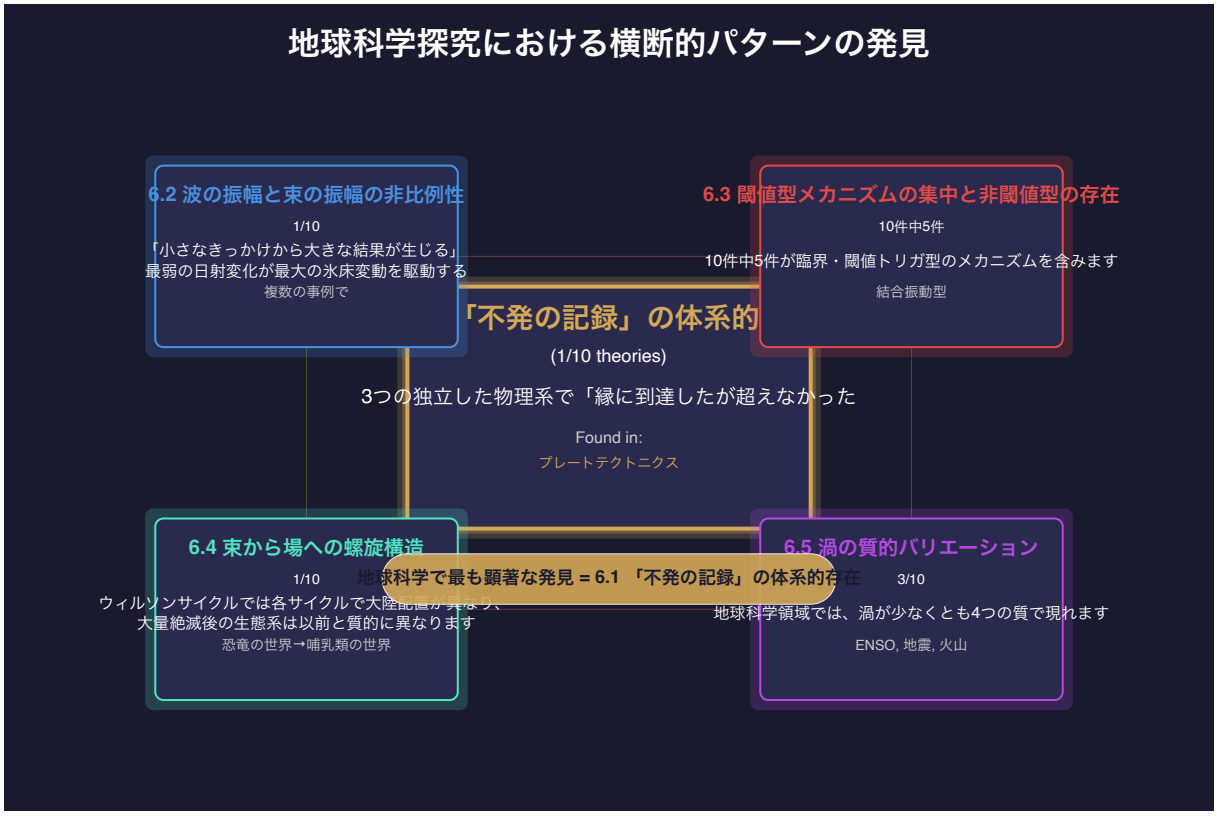
- 5


10地球科学理論と5段階モデルの対応関係マトリクス
場
波
縁
渦
束
1. プレートテクトニクス（ウィル
2. ENSO（大気海洋結合振動）
3. ミランコビッチサイクル（軌道強制）
4. 地震発生メカニズム（弾性反跳
5. 火山噴火ダイナミクス
6. シーケンス層序学
7. 大量絶滅と回復
8. 風化・土壌形成
9. 地球磁場ダイナモと磁場逆転
10. 惑星分化（コア-マントル-地殻
強い対応 (Strong)
部分的 (Partial)
条件付き/弱い (Conditional/Weak)

2: Domain — ×5

3: ENSO

- 2 10 Philander 1983 Bjerknes Bjerknes 1969 Cane and Zebiak 1985 Suarez and Schopf 1988
- Bjerknes feedback
- ENSO 5
- Walker Kelvin Rossby SST El Nino La Nina
- ENSO 5
- ENSO 5



3: Domain —

- ENSO
- failed rift
- ENSO
-

-
- 100kyr
- ENSO
- 5

- 5
-